Orthic Ferralsols:

Orthic Ferralsols are deep, well-drained tropical soils with a reddish to yellowish color, clay-rich subsurface, and strongly weathered, physically stable structure.

They are named “Orthic” because they have relatively higher base saturation (>35%) than Acric Ferralsols, making them less acidic and more fertile. These soils typically occur on undulating to gently sloping terrain in humid tropical regions and are suitable for both plantation and food crops.

Drainage:

- **Wet season:** Well-drained; temporary ponding may occur in micro-depressions.
- **Dry season:** Soil remains friable and workable; moderate moisture retention supports crops.
- **Landscape:** Common on undulating to gently sloping tropical uplands; rarely found in depressions.

Depth:

- Very deep (>150 cm); rooting depth generally unrestricted due to stable, porous structure.

Workability:

- **When wet:** Plastic and sticky but manageable; does not smear like Vertisols.
- **When dry:** Firm, friable, and easy to till; seedbed preparation straightforward.

Nutrient Richness & Cation Exchange Capacity (CEC):

- CEC: Moderate to high (15–35 cmol(+)/kg).
- Base saturation: Moderate (>35%), higher than Acric Ferralsols.
- Nitrogen and potassium moderate; organic matter low to moderate.
- Phosphorus may be limited due to fixation.
- Iron and aluminum oxides dominate; some micronutrients sufficient for crop growth.

pH:

- Slightly acidic to near neutral (5.0–6.5).

Management:

- Add organic matter (manure, compost, crop residues) to improve fertility, structure, and water retention.
- Apply phosphorus fertilizers carefully (rock phosphate + small doses of superphosphate).
- Liming may be required if pH drops below optimal levels.
- Maintain vegetation cover or cover crops to reduce erosion.
- Avoid heavy machinery when the soil is wet.
- Suitable for cocoa, coffee, oil palm, rubber, and properly fertilized food crops.